

Identifying novel therapeutic targets of natural compounds in traditional Chinese medicine herbs with hypergraph representation learning

Yantong Qiao^{1,2}, Lun Hu^{1,*}, Jun Zhang¹, Pengwei Hu¹, Xin Luo²

¹Xinjiang Technical Institute of Physics and Chemistry, Chinese Academy of Sciences, No. 40-1 Beijing Road, 830011 Urumqi, China

²College of Computer and Information Science, Southwest University, No. 2 Tiansheng Road, 400715 Chongqing, China

*Corresponding author. Xinjiang Technical Institute of Physics and Chemistry, Chinese Academy of Sciences, No. 40-1 Beijing Road, 830011 Urumqi, China.

E-mail: hulun@ms.xjb.ac.cn

Abstract

Traditional Chinese medicine (TCM), with its roots in centuries of clinical practice, has established itself as an effective therapeutic system that involves a diverse range of herbal plants. Despite its proven efficacy, the intricate relationships between herbal multi-component preparations and multi-target therapies present challenges for systematic study, thereby limiting its broader application in managing chronic diseases. In this work, we aim to identify novel therapeutic targets of natural compounds found in TCM herbs by leveraging advanced hypergraph representation learning techniques. Following the multi-component, multi-target pharmacological mechanisms, we first construct two hypergraphs to represent herb–compound and disease–target interactions, respectively. The connection between these hypergraphs is established through compound–target associations. A convolutional operator is then employed to capture the high-order correlations between compound (or target) nodes and herb (or disease) hyperedges within each hypergraph. Furthermore, we incorporate the PageRank algorithm and a multi-head attention mechanism to enhance the representation capabilities of node embeddings. By integrating these methods, our model is able to accurately identify novel therapeutic targets of natural compounds in TCM herbs in an end-to-end manner. Extensive experiments conducted on three benchmark datasets demonstrate the superior performance of our model when compared with several state-of-the-art approaches. Furthermore, case studies on two natural compounds, coumarin and progesterone, reveal that 7 and 8 out of the Top-10 identified targets, respectively, have been validated through literature review. These results highlight the effectiveness of our model in discovering new therapeutic targets for natural compounds in TCM.

Keywords: compound–target interaction prediction; hypergraph representation learning; multi-compound multi-target mechanisms; Chinese tradition medicine

Introduction

Chinese herbal medicine (CHM) is a foundational element of traditional Chinese medicine (TCM), with a rich history and widespread applications across various medical practices [1]. Through centuries of practice and generational knowledge transfer, it has developed a unique and comprehensive theoretical system, along with extensive experience in medicinal use. Unlike modern drugs, which are typically based on single compounds, CHM comprises multiple active components. This complexity allows it to interact with various disease targets, presenting the potential to treat multiple conditions simultaneously. For instance, Chenpi (dried citrus peel) and Danggui (Angelica sinensis) are traditional Chinese herbs frequently included in medicinal formulations for treating a range of diseases. Chenpi contains active compounds, such as hesperidin, apigenin, and β -sitosterol, which target proteins, including Uroporphyrinogen Decarboxylase (UROD), Fatty Acid Binding Protein 1 (FABP1), and Carnitine palmitoyltransferase II (CPT2), respectively. These interactions make Chenpi effective for treating diseases like Alzheimer, hypertension, inflammation, and

other health issues [2]. Similarly, Danggui is rich in bioactive compounds like decanal, sodium tungstate, and tomatine, which act on targets, such as Butyrylcholinesterase (BCHE), Choline Acetyltransferase (ChAT), and Retinol-binding Protein 4 (RBP4), respectively. This activity makes it effective for treating periodontitis, blood system disorders, and coronary heart disease [3].

However, the multi-compound, multi-target (MCMT) nature of CHM creates complex interactions with disease targets that remain only partially understood. This gap in knowledge limits our ability to fully harness its therapeutic potential [4]. Existing predictive models often fail to capture these intricate mechanisms, limiting the accuracy of identifying novel therapeutic targets of natural compounds in CHM. Addressing these complexities is crucial, though it requires further advances to improve the prediction performance of compound–target interactions (CTIs) and maximize the therapeutic efficacy of CHM. In recent years, the systematic approach of network pharmacology has gained significant attention in the study of TCM [5]. This approach provides a novel paradigm to explore and visualize the intricate interaction

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networks of TCM in addressing multifactorial diseases [6]. By constructing heterogeneous information networks that encompass various relationships, such as herb–compound, compound–target, and target–disease associations, it opens new avenues for more accurate CTI predictions.

Integrating network pharmacology with traditional herbal databases is crucial for predicting CTIs in clinical herbal applications and formulation development [7]. Given that both herbal compounds and drugs are small molecules, researchers often treat herbal compounds as drugs, reframing the CTI prediction problem as a challenging drug–target interaction (DTI) prediction task. Numerous computational methods have been developed for DTI prediction, broadly categorized into three main types: machine learning (ML), deep learning (DL), and graph neural network (GNN)-based methods. Each of these methods presents distinct advantages while also encountering unique challenges in effectively capturing the complex MCMT nature of TCM.

ML-based methods employ algorithms to model interactions between entities, such as drugs and targets, often by extracting features from biological data [8]. These methods typically utilize traditional algorithms like support vector machines (SVMs), random forests, and decision trees to predict DTIs. A common characteristic of ML approaches is their reliance on handcrafted features derived from biological datasets. For instance, Shi et al. [9] combine Lasso regression with a random forest model, leveraging evolutionary information, and chemical structure features to enhance prediction accuracy. Similarly, Chen et al. [10] apply a random forest algorithm to utilize its ensemble learning capabilities, particularly suited for handling high-dimensional biomedical data. Song et al. [11] adopt a similarity-based SVM model that employs kernel functions to map DTIs into a higher-dimensional feature space, improving prediction performance. Keum and Nam [12] introduce the SELF-BLM model, which integrates a semi-supervised SVM with self-training to incorporate unlabeled data, thereby enriching the predictive process. Finally, Sorkhi et al. [13] propose a hybrid model combining decision trees with SVMs, where decision trees capture local features and SVMs classify DTIs based on these localized characteristics.

DL-based methods extend the capabilities of ML-based methods by learning complex, high-level representations of drugs and targets. These methods are able to capture local and global features from protein sequences and drug molecular structures using advanced neural network architectures [14–16]. By employing multi-layer neural networks, they can automatically extract intricate features, making them particularly effective for analyzing complex molecular data, and identifying DTIs. For instance, DeepDTA [17] employs convolutional neural networks (CNNs) to model 1D representations of protein sequences and drug compounds, effectively capturing local features for DTI prediction. Similarly, HyperAttentionDTI [18] integrates CNNs with attention mechanisms in an end-to-end bionic model. It uses deep CNNs to learn feature matrices for drugs and proteins, and attention vectors to analyze non-covalent intermolecular interactions between atoms and amino acids, enhancing DTI prediction. The CoaDTI model [19] leverages a co-attention mechanism to model drug–protein interactions. It transforms a drug's SMILES sequence into radius subgraphs for embedding via GraphSage while extracting protein features using a Transformer encoder. Self-attention encodes both drug and protein features, and their interactions are modeled through DPA and PDA layers. A fully connected layer then maps these features to interaction probabilities for prediction. MCL-DTI [20] employs multi-head self-attention and

cross-attention mechanisms to generate embedding representations for drugs and targets. These embeddings interact to form a combined feature map, which is subsequently passed through a classifier to predict DTIs. Lastly, PerceiverCPI [21] utilizes a cross-attention mechanism to enable deep interaction between drug and target features. By leveraging rich information from extended drug SMILES sequences, PerceiverCPI predicts DTIs with high precision, making it an effective tool for understanding DTIs.

GNN-based methods advance the field by representing DTIs as graph structures, which effectively capture intricate relationships and dependencies within the data [22–24]. These methods model drugs and targets as nodes and interactions as edges, enabling a natural and intuitive representation of complex biological systems. GraphormerDTI [25] employs a Graph Transformer neural network to predict DTIs by encoding molecular structures and their interactions with high precision. Similarly, GandTI [26] integrates a GNN with an attention mechanism to predict both DTIs and binding affinities in an efficient end-to-end framework. HGNNLDA [27] applies a hypergraph neural network to predict lncRNA–drug sensitivity. The model begins by constructing a bipartite graph of lncRNA–drug sensitivity associations, which then extends into separate hypergraphs for lncRNAs and drugs. These hypergraphs are used to generate embeddings for lncRNAs and drugs, and the sensitivity levels between them are calculated using inner product computation. DrugBAN [28] is a multimodal model designed for drug–protein interaction prediction. It extracts drug features using a graph convolutional network (GCN) and protein features with a CNN. These features are fused in a bilinear attention network to form a joint representation, which is passed through a fully connected layer for interaction probability prediction. Additionally, DrugBAN incorporates a CDAN to enhance generalization across different data domains. Lastly, BINDTI [29] is an end-to-end DTI recognition model that employs a bidirectional intention network. Drug features are encoded using a GCN, while target features are represented with a mixed ACmix model combining self-attention and convolution. The bidirectional intention network integrates these features, and the resulting fused representation is classified using a multi-layer perceptron to predict DTIs.

Despite notable advances, existing methods mentioned earlier face significant challenges in modeling the complex MCMT characteristic of TCM. These challenges stem from several limitations. First, most current methods primarily address single-component, single-target interactions, making them insufficient for capturing the intricate MCMT dynamics typical of TCM. Second, they often neglect the synergistic effects among multiple components—a hallmark of traditional herbal therapies. Third, the scarcity and heterogeneity of high-quality experimental data pose significant barriers for accurate prediction.

To address these limitations and advance our understanding of MCMT interactions between herbal compounds and disease targets, we propose a novel computational model called hypergraph representation learning for compound–target interaction (HDCTI) prediction. As illustrated in Fig. 1, HDCTI begins by generating initial embeddings for herbal compounds and disease targets through sampling from truncated normal distributions, iteratively refining these embeddings during training. The model constructs two distinct hypergraphs: one representing relationships among herbs and their compounds, and another capturing associations between diseases and their target proteins.

Unlike our previous model GHDA [30], which was designed to predict associations between herbal plants and diseases by optimizing hyperedge embeddings, HDCTI shifts the focus to

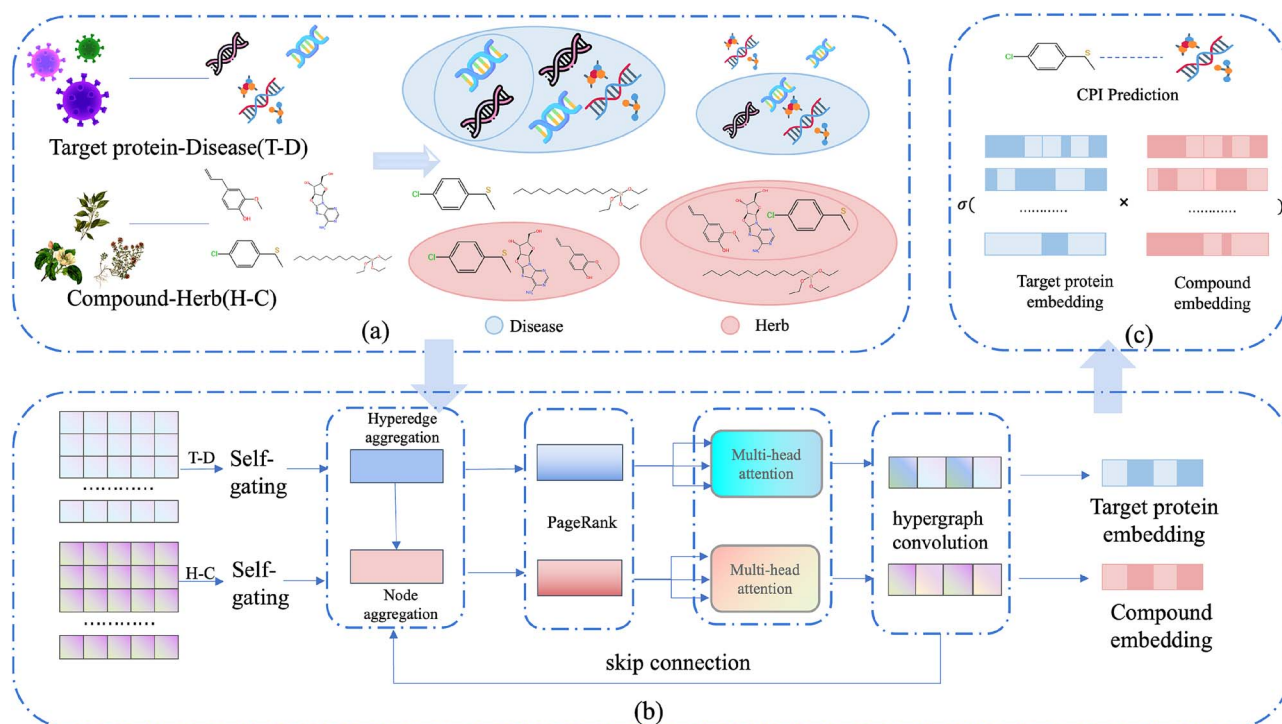


Figure 1. Overview of the model. (a) hypergraph construction, (b) targets and compounds embeddings acquisition, and (c) potential CTI prediction. Alt Text: A diagram showing the hypergraph-based model structure, including graph construction, embedding learning, and interaction prediction.

identifying therapeutic targets of natural compounds. The key differences between our current work and our previous study can be summarized in two aspects. First, although both algorithms, i.e. HDCTI in this work and HGHDA in our previous study, utilize a hypergraph structure to model complex multi-component, multi-target (MCMT) relationships in TCM, they address different problems. HGHDA was designed to predict associations between hyperedges, specifically herbal plants and diseases, whereas HDCTI focuses on identifying therapeutic targets of natural compounds, with both targets and compounds represented as nodes in the hypergraph. While HGHDA can generate embeddings for targets and compounds, its primary objective is to optimize hyperedge embeddings, limiting its ability to capture target-compound relationships and reveal their intricate associations. Second, in hypergraph convolutional networks, node embeddings are learned by aggregating information from both pairwise edges and higher-order relationships encoded in hyperedges. However, effectively capturing global structural significance and adaptive neighborhood dependencies in hypergraphs remains a challenge. To address this, we integrate the PageRank algorithm with a multi-head attention mechanism, enhancing node embeddings through global influence propagation and context-aware feature aggregation. Experimental results demonstrate that HDCTI significantly outperforms HGHDA and other existing prediction models across multiple datasets. Its robustness is further validated through case studies involving compounds such as coumarin and progesterone, where it achieves notable predictive success.

Materials and methods

Datasets

To evaluate the performance of HDCTI in predicting CTIs, we collect the data of herb-compound interactions, CTIs, and

Table 1. Details for TCM-Suite, TCMSP, and SymMap2.0 Datasets

	Number		
	TCM-Suite	TCMSP	SymMap2.0
Herb	1009	502	697
Compound	1193	13 716	27 277
Protein target	7258	1749	18 192
Disease	11 071	322	12 690
Herb-compound	6496	33 933	85 172
Compound-protein	43 669	56 169	38 043
Protein-disease	44 170	173 639	196 110

disease-target interactions from three publicly available CTM databases, i.e. TCM-Suite [31], TCMSP [32], and Symmap2.0 [33], respectively, to construct hypergraphs. Notably, the three datasets exhibit limited overlap, with herb and disease nodes sharing only a 6% direct overlap rate across all datasets, while compound-target relationship pairs show even lower redundancy, with <3% repetition. These findings indicate that each dataset used in our experiments is significantly distinct from the others due to its low duplication rate. Detailed data descriptions are presented in Table 1, where “Compound-protein” denotes the number of positive samples used in the experiments. To train and evaluate our model, negative samples are randomly selected from compound-target pairs that are not present in the respective datasets. To mitigate potential bias caused by class imbalance, we ensure that the number of negative samples is equal to that of positive samples.

Hypergraph construction

Through the known interactions between herbs-compounds and targets-diseases, we can establish two interaction matrices,

denoted as A_{hc} and A_{td} , and use them as inputs to construct hypergraphs. A hypergraph is a generalization of a graph. In a graph, an edge can connect two vertices. In a hypergraph, however, each edge can connect any number of vertices, such edges are called hyperedges. This special mathematical structure allows hypergraphs to represent more complex relationships in a more flexible manner. In our work, we will construct two hypergraphs, namely the herb-compound hypergraph and the target-disease hypergraph. In the herb-compound hypergraph, herbal compounds are nodes of the hypergraph, and herbs are hyperedges. In the target-disease hypergraph, targets affected by diseases are nodes of the hypergraph, and diseases are hyperedges.

The hypergraph is represented as $G = (V, \epsilon, A)$, where $V = \{v_i\}$ denotes the set of nodes, with a length of m ; $\epsilon = \{e_j\}$ represents the set of hyperedges, with a length of n ; $A \in \{0, 1\}^{m \times n}$ is an indicator matrix used to denote the existence or the absence of nodes and hyperedges in the hypergraph.

Each element $a(v_i, e_j)$ in A indicates whether node v_i is part of hyperedge e_j . Specifically, if $a(v_i, e_j) = 1$, it means node v_i is part of hyperedge e_j ; if $a(v_i, e_j) = 0$, it means node v_i is not part of hyperedge e_j . Thus, the indicator matrix A encodes the relationship between nodes and hyperedges in G .

Based on the definition of G , we can construct two hypergraphs, denoted as $G_{hc} = \{V_c, \epsilon_c, A_{hc}\}$ and $G_{td} = \{V_t, \epsilon_t, A_{td}\}$. Through the form of hypergraphs, HDCTI represents the complex relationship between herbal compounds and disease targets. By utilizing hypergraphs, HDCTI can comprehensively consider the interactions among different compounds and the diverse connections between targets and diseases. This enables HDCTI to effectively consider the MCMT characteristics between CHM and compounds, thereby improving the accuracy of CTI prediction.

Hypergraph representation learning

The HDCTI model utilizes hypergraph representation learning, which can be divided into two main components: a self-gating mechanism and hypergraph convolution. The hypergraph convolution part is further broken down into five steps: hyperedge aggregation, node aggregation, PageRank integration, a multi-head attention mechanism, and hypergraph convolution. These components collectively enhance the model's capability to capture intricate relationships between compounds and targets, improving prediction accuracy.

Self-gating

In this step, HDCTI introduces a self-gating mechanism, with the purpose of selectively controlling the flow of information and enhancing the representation capabilities for herb compounds and disease targets. Specifically, we create two gating units in two hypergraphs, respectively, to regulate the herb compounds and disease targets. Assuming that E_c^{ini} and E_t^{ini} denote the initial embeddings of herb compounds and disease targets by following a truncated normal distribution, the specific self-gating operations are as follows:

$$E_c^0 = E_c^{ini} \cdot \sigma(E_c^{ini} W_{gc} + b_{gc}), \quad (1)$$

$$E_t^0 = E_t^{ini} \cdot \sigma(E_t^{ini} W_{gt} + b_{gt}), \quad (2)$$

where σ is the sigmoid function which generates a gating signal between 0 and 1, controlling the flow of information. The self-gating mechanism enables the model to filter and emphasize relevant information dynamically, allowing for a more nuanced

integration of diverse features. W_{gc} and W_{gt} represent the weight matrices of gate units in the two channels, while b_{gc} and b_{gt} are bias terms. E_c^0 and E_t^0 are the embeddings of compounds and targets after gating, and they are the input for hypergraph convolution. This gating process ensures that the model can adaptively regulate the contribution of each feature, thus better capturing the characteristics of both herb compounds and disease targets.

Hypergraph convolution

Hyperedge convolution In this step, HDCTI aggregates information from compound node V_c to update the herb hyperedge embedding E_h . For each hyperedge, the model gathers information from the nodes within it and combines this information to form an embedding for the hyperedge. Such embeddings can capture higher-order patterns and interactions among nodes within the same hyperedge. To perform node-hyperedge transformation, hyperedge aggregation is defined as:

$$E_h^l = D_h^{-1} A_{hc}^T E_c^{(l-1)}, \quad (3)$$

where D_h is the edge degree matrix of A_{hc} , E_h^l is the embedding matrix of herb hyperedges at l th layer, and E_c^l is the embedding matrix of compound nodes at l th layer. With (3), we can achieve information propagation from nodes to hyperedges through the operation defined by $A_{hc}^T E_c^l$.

Node aggregation Given E_h^l , HDCTI then performs node aggregation operations to update the embeddings of compound nodes at l th layer, denoted as E_c^l . The model updates the compound node embeddings by aggregating information from the hyperedge embeddings E_h^l using the node's adjacency relations with hyperedges. Specifically, the aggregation operation is defined as:

$$E_c^l = D_c^{-1} A_{hc} E_h^l, \quad (4)$$

where D_c is the node degree matrix of A_{hc} . Furthermore, according to the above equations, by the operations $A_{hc} E_h^l$, we aggregate hyperedge information into node embeddings.

PageRank integration In this step, we apply the PageRank algorithm to the herb-compound hypergraph, focusing on capturing the global importance of each node [34]. The goal is to quantify how influential a node is within the entire network, which helps the model prioritize nodes that play a more central role in the hypergraph's structure. By calculating the PageRank values of nodes, we enhance the node embeddings with a global perspective, emphasizing key nodes that contribute significantly to the information flow across the hypergraph. To indicate the relative importance of node v , its PageRank value is defined as below.

$$PR(v) = \frac{1 - \alpha}{N} + \alpha \sum_{u \in M(v)} \frac{PR(u)}{L(u)}, \quad (5)$$

where $M(v)$ is the set of neighboring nodes of v , and $L(u)$ is the degree of node u . Assuming that $v_{ci} \in V_c$ is an arbitrary compound node, its corresponding embedding in E_c is E_{ci} . We then update E_{ci} with (6).

$$E_{ci} = E_{ci} \times PR(v_i) \quad (6)$$

By updating all the embeddings in E_c^l with (6), we ensure that the model leverages the global importance in the hierarchical structure of G_{hc} , allowing it to focus on more significant nodes and

improve overall prediction accuracy. In this way, the compound embeddings not only retain the original node features but also integrate the global significance of the nodes within the network, ensuring that the model can leverage this global information to optimize its representation capacity during training.

Multi-head attention In this step, HDCTI introduces the multi-head attention mechanism to enhance the model's ability to capture local interactions between nodes by focusing on different representation subspaces simultaneously. This allows the model to pay attention to various aspects of node relationships at finer granularities, enriching the understanding of local context.

For the l th layer input embeddings E_c^l , we compute the queries $Q_c^{(lh)}$, keys $K_c^{(lh)}$, and values $U_c^{(lh)}$:

$$Q_c^{(lh)} = E_c^l W_{qc}^{(lh)}, \quad K_c^{(lh)} = E_c^l W_{kc}^{(lh)}, \quad U_c^{(lh)} = E_c^l W_{uc}^{(lh)} \quad (7)$$

where h denotes the h th attention head; $W_{qc}^{(lh)}$, $W_{kc}^{(lh)}$, and $W_{uc}^{(lh)}$ are trainable weight matrices. This approach allows each attention head to extract features from different subspaces, capturing various local relationships among nodes.

The local attention score for the h th head at the l th layer is calculated as follows:

$$S_c^{(lh)} = \sigma \left(\frac{Q_c^{(lh)} (K_c^{(lh)})^T}{\sqrt{d_k}} \right) \cdot U_c^{(lh)}, \quad (8)$$

where d_k is the dimensionality of the key vectors in $K_c^{(lh)}$, and σ denotes the softmax function.

These local attention scores help the model focus on specific relationships in the hypergraph, capturing detailed contextual information. The final output S_c^l , derived by combining the outputs from multiple attention heads with (9), integrates information from various attention heads to provide a richer and more nuanced representation of the compound relationships

$$S_c^l = [S_c^{(l1)}, S_c^{(l2)}, \dots, S_c^{(lh)}]. \quad (9)$$

By incorporating multi-head attention, the model enhances its ability to recognize nuanced, local interactions, complementing the global node importance derived from the PageRank algorithm. Together, these mechanisms ensure that both local context and global significance are effectively captured in the compound-target prediction process.

Hypergraph convolution Combining the operations of hyper-edge aggregation and node aggregation described above, we can define hypergraph convolution as:

$$E_c^l = D_c^{-1} A_{hc} D_h^{-1} A_{hc}^T S_c^l, \quad (10)$$

where S_c^l is the output of multi-head attention obtained at layer l .

During the HDCTI learning process, combining multi-level aggregations together may lead to the problem of feature information dilution. To address this issue, the model integrates skip connections similar to ResNet to prevent information dilution across different layers. This skip connection mechanism enables HDCTI to directly access low-level feature information. Therefore, the final hypergraph convolution should be defined as follows:

$$E_c^l = \sigma \left(\left(D_c^{-1} A_{hc} D_h^{-1} A_{hc}^T S_c^l \right) \theta^l + E_c^{(l-1)} \right), \quad (11)$$

where $\sigma(\cdot)$ represents a non-linear activation function, such as the ReLU function and the LeakyReLU function. θ^l is a trainable parameter. This allows the model to simultaneously consider the original features and aggregated embeddings, thereby generating the final representation of herb compounds.

HDCTI performs hypergraph convolution via hyperedge aggregation, node aggregation, and multi-head attention, enabling effective utilization of the hypergraph structure to capture and propagate information. The embedding of target E_t can be obtained by the same way. Considering the complex MCMT relationships between herb compounds and disease targets, this allows the model to learn meaningful and informative embeddings of components and targets, which can be used for predicting interactions between compounds and targets.

Compound-target interaction prediction

Through the above steps, we can obtain the final embeddings E_c and E_t for herb compound and disease target, respectively. The final embeddings are:

$$E_c = \frac{1}{l_{\max}} \sum_{l=1}^{l_{\max}} E_c^l, \quad (12)$$

$$E_t = \frac{1}{l_{\max}} \sum_{l=1}^{l_{\max}} E_t^l, \quad (13)$$

where E_c^l and E_t^l are the embeddings obtained from hypergraph convolution at layer l , and $l_{\max}$ is the maximum number of layers.

With (12) and (13), we can calculate the prediction score between $v_{ci} \in V_c$ and $v_{tj} \in V_t$ as:

$$\hat{p}_{ij} = \sigma(E_{ci} E_{tj}^T), \quad (14)$$

where σ denotes the sigmoid function, E_{ci} and E_{tj} are the embeddings of v_{ci} and v_{tj} in E_c and E_t , respectively. To optimize the model, we use the cross-entropy loss function to calculate the loss during the training process:

$$L_{CE} = \sum_{i,j} \left[-A_{ct}^{ij} \ln(\hat{p}_{ij}) - (1 - A_{ct}^{ij}) \ln(1 - \hat{p}_{ij}) \right] + \lambda \|\Theta\|_2^2, \quad (15)$$

where Θ represents the trainable parameters of the model, A_{ct}^{ij} is equal to 1 if v_{ci} and v_{tj} are associated and 0 otherwise. We introduce a regularization term $\lambda \|\Theta\|_2^2$ to calculate the L2 regularization of Θ . This approach can reduce generalization error, where λ is a hyperparameter that adjusts the impact of Θ on model optimization.

Experiments

Evaluation metrics

To evaluate the performance of HDCTI, we conducted extensive experiments using five-fold cross-validation. Both positive and negative samples were rigorously constructed as described in Section 2 Datasets. Specifically, negative samples were randomly selected from unknown compound-target pairs that are not recorded in TCM-Suite, TCMSp, or Symmap2.0 databases, ensuring no overlap with positive interactions. In the five-fold cross-validation scheme, all positive and negative samples are randomly divided into five equal-sized folds. Each fold is alternately used as the test set, while the remaining four folds serve

as the training set. Due to the random partitioning, some targets in the test set may not necessarily appear in the training set, which ensures a fair evaluation of the model's ability to predict unseen associations. Additionally, in our hypergraph-based model, diseases are utilized to define hyperedges, each comprising related target nodes. The primary purpose of incorporating diseases is to enhance the learning of target embeddings through hypergraph convolution, rather than to serve as predictive variables in the training or evaluation process. In our experiments, the five-fold cross-validation was repeated five times, and the average performance was reported for comparative analysis. We used several independent evaluation metrics to measure the predictive performance of the model.

First, we employed the area under the ROC curve (AUC) and the area under the precision-recall curve (AUPR), which are effective indicators for evaluating the performance of binary classification models. AUC assesses the overall predictive performance of the model at different thresholds; AUPR is particularly valuable when dealing with imbalanced datasets. Both metrics range from 0 to 1, with values closer to 1 indicating better predictive performance of the model.

Next, we used Recall, Precision, and F1-score, where the F1-score is the harmonic mean of Precision and Recall. Their calculations are as follows:

$$\text{Recall} = \frac{TP}{FN + TP}, \quad (16)$$

$$\text{Precision} = \frac{TP}{FP + TP}, \quad (17)$$

$$\text{F1-score} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}, \quad (18)$$

where TP represents the number of true positive samples correctly predicted; FN represents the number of positive samples incorrectly predicted; FP represents the number of negative samples incorrectly predicted. The F1-score ranges from 0 to 1, with values closer to 1 indicating higher model accuracy. An F1-score of 1 signifies that both precision and recall have reached a perfect state. Conversely, an F1-score of 0 indicates the worst model performance.

Baseline algorithms

To validate the effectiveness of HDCTI, we compared its performance with several state-of-the-art models in the field of DTI prediction. These well-regarded models were selected as benchmarks to comprehensively evaluate HDCTI, including HyperAttentionDTI [18], CoaDTI [19], HGNNLDA [27], DrugBAN [28], MCL-DTI [20], PerceiverCPI [21], BINDTI [29], and HGHDA [30]. Of these, HyperAttentionDTI, CoaDTI, MCL-DTI, and PerceiverCPI are DL-based, while the remaining models are GNN-based.

Comparative analysis with baseline algorithms

This section provides a comprehensive comparison of the HDCTI model's ability to predict CTIs with several baseline methods.

The results of the comparative experiments, as shown in Table 2 and Fig. 2, comprehensively demonstrate the superior performance and robustness of HDCTI across multiple datasets. HDCTI consistently outperforms other methods in key performance metrics such as AUC, AUPR, Recall, Precision, and F1-score. Moreover, it maintains stable performance under different data

splits and cross-validation folds, with low standard deviations indicating minimal variance. These results highlight the strong generalization capability and reliability of the model in predicting interactions.

In the comparative analysis, HDCTI leads across the three datasets. Notably, on the SymMap2.0 dataset, HDCTI outperformed CoaDTI by >30%. The superior performance of HDCTI may be attributed to its unique structure, which effectively captures high-order information using a multi-head attention mechanism, PageRank integration, and hypergraph convolution, thereby preserving the fundamental characteristics of the compounds and targets.

Regarding HyperAttentionDTI, DrugBAN, and BINDTI, these three models all utilized convolution and attention mechanisms. However, from the perspective of evaluation indicators, although the AUC index of HyperAttention DTI performs well on all three datasets and is better than the other two models. But especially on the SymMap 2.0 dataset, the precision and AUPR indices are clearly poor. Particularly on the SymMap2.0 dataset, there were significant differences in Precision and AUPR indices. The performance differences, despite all using convolution and attention mechanisms, might be due to HDCTI's use of a hypergraph structure, which can better capture high-order information and prevent information loss.

For HGNNLDA, even though this model uses a hypergraph structure similar to ours, its performance significantly lags behind HDCTI. This could be related to the hypergraph structure employed by HGNNLDA, which might not adequately address the complex multi-target, multi-component mechanisms in CPI discovery.

Regarding CoaDTI, MCL-DTI, and PerceiverCPI, all three models are based on attention mechanism networks. Despite their good performance on the AUC metric, they exhibit relatively poor recall, precision, F1-score, and AUPR on the TCMSp and SymMap2.0 datasets. This indicates insufficient performance in predicting positive samples, likely due to the models failing to capture enough feature information, leading to a lack of accuracy, and recall in predictions. In contrast, HDCTI effectively captures unique features by obtaining embeddings of components and targets through two independent hypergraphs, demonstrating its effectiveness.

As for HGHDA, it demonstrates certain advantages in predicting associations between herbs and diseases, but its primary focus on optimizing hyperedge embeddings limits its ability to effectively capture the complex relationships between targets and compounds. This design constraint affects its performance in identifying therapeutic targets for natural compounds. In contrast, HDCTI integrates the PageRank algorithm and multi-head attention mechanism, which allows for better capture of adaptive dependencies and global influences between nodes, significantly improving performance. Experimental results show that HDCTI outperforms HGHDA by 24.3%, 24.5%, and 25.7% in terms of AUC, AUPR, and F1-score, respectively, demonstrating greater robustness and accuracy.

To further ensure that the presence of the same diseases in both the training and test sets does not lead to data leakage, we designed a rigorous control experiment. Specifically, we first randomly selected 10 disease nodes from the entire disease set and identified all protein targets associated with these diseases. Subsequently, all CTIs involving these targets were designated as the test set, while the remaining interactions formed the training set. This disease-aware validation strategy ensures that no protein targets linked to the same disease appear in both the training and

Table 2. Performance comparison of different methods on three datasets

Dataset	Method	AUC	AUPR	Recall	Precision	F1-score
TCM-Suite	HyperAttentionDTI	0.9870(±0.0008)	0.9708(±0.0015)	0.9534(±0.0072)	0.9275(±0.0025)	0.9401(±0.0045)
	CoaDTI	0.8758(±0)	0.7751(±0)	0.5997(±0)	0.9124(±0)	0.7237(±0)
	HGNNLDA	0.8252(±0.0024)	0.7530(±0.0049)	0.8862(±0.0026)	0.4352(±0.0029)	0.5837(±0.0028)
	DrugBAN	0.9799(±0.0042)	0.9429(±0.0033)	0.8745(±0.0012)	0.9444(±0.0084)	0.9081(±0.0032)
	MCL-DTI	0.9741(±0.0053)	0.9405(±0.0084)	0.8767(±0.0052)	0.9759(±0.0132)	0.9236(±0.0075)
	PerceiverCPI	0.9862(±0.0004)	0.9581(±0.0005)	0.8809(±0.0045)	0.9834(±0.0064)	0.9293(±0.0022)
	BINDTI	0.9603(±0.0021)	0.9029(±0.0101)	0.8487(±0.0047)	0.9698(±0.0303)	0.9049(±0.0107)
	HGHDA	0.6282(±0.0192)	0.6289(±0.0104)	0.7180(±0.0141)	0.5734(±0.0350)	0.6366(±0.0161)
	Ours	0.9917(±0.0005)	0.9934(±0.0002)	0.9475(±0.0014)	0.9887(±0.0014)	0.9677(±0.0008)
TCMSP	HyperAttentionDTI	0.9802(±0.0042)	0.8550(±0.0204)	0.8346(±0.0037)	0.6296(±0.0323)	0.7172(±0.0196)
	CoaDTI	0.6666(±0)	0.2474(±0)	0.0305 (±0)	0.4067(±0)	0.0568(±0)
	HGNNLDA	0.9032(±0.0021)	0.9130(±0.0017)	0.9369(±0.0033)	0.5608(±0.0026)	0.7016(±0.0023)
	DrugBAN	0.9733(±0.0013)	0.7280(±0.0046)	0.5588(±0.0028)	0.7503(±0.0073)	0.6406(±0.0044)
	MCL-DTI	0.9254(±0.0086)	0.6638(±0.0175)	0.6337(±0.0127)	0.6724(±0.0066)	0.6524(±0.0082)
	PerceiverCPI	0.9759(±0.0003)	0.8238(±0.0020)	0.7049(±0.0060)	0.7751(±0.0013)	0.7383(±0.0027)
	BINDTI	0.9720(±0.0004)	0.8649(±0.0019)	0.7294(±0.0109)	0.8297(±0.0002)	0.7762(±0.0061)
	HGHDA	0.9376(±0.0035)	0.9239(±0.0047)	0.9156(±0.0055)	0.8310(±0.0074)	0.8712(±0.0033)
	Ours	0.9890(±0.0005)	0.9867(±0.0007)	0.9781(±0.0011)	0.9439(±0.0016)	0.9607(±0.0011)
SymMap2.0	HyperAttentionDTI	0.9478(±0.0010)	0.4344(±0.0052)	0.2589(±0.0048)	0.5475(±0.0241)	0.3713(±0.0006)
	CoaDTI	0.6276(±0)	0.2445(±0)	0.0121 (±0)	0.3077 (±0)	0.0233 (±0)
	HGNNLDA	0.8692(±0.0017)	0.8839(±0.0015)	0.9104(±0.0039)	0.5526(±0.0022)	0.6878(±0.0019)
	DrugBAN	0.8032(±0.0485)	0.1900(±0.0125)	0.2087(±0.0596)	0.4055(±0.0966)	0.2599(±0.0522)
	MCL-DTI	0.8369(±0.0030)	0.2525(±0.0096)	0.2702(±0.0152)	0.3868(±0.0239)	0.3181(±0.0186)
	PerceiverCPI	0.9540(±0.0004)	0.4256(±0.0020)	0.1588(±0.0217)	0.6993(±0.0250)	0.2577(±0.0266)
	BINDTI	0.9338(±0.0015)	0.4679(±0.0013)	0.2020(±0.0007)	0.7289(±0.0027)	0.3164(±0.0012)
	HGHDA	0.8722(±0.0059)	0.8702(±0.0063)	0.7533(±0.0373)	0.8192(±0.0130)	0.7841(±0.0160)
	Ours	0.9632(±0.0010)	0.9610(±0.0015)	0.9180(±0.0019)	0.8979(±0.0028)	0.9078(±0.0017)

Note: Best results are bolded.

test sets, thereby effectively mitigating the risk of disease-related information leakage during model evaluation.

To ensure fairness and consistency, we have applied the disease-aware splitting protocol uniformly across all methods, including the baselines. Specifically, the same set of 10 disease nodes and their associated protein targets were excluded from the training sets of all models. This consistent setting prevents disease-related information leakage and guarantees that all models have been fairly evaluated under the same stringent conditions.

In Table 3, a performance drop is observed across all models, with the decline being particularly notable among the baseline methods. This phenomenon primarily stems from the disease-aware validation setting, which strictly prohibits any protein targets associated with the same disease from appearing in both the training and test sets. As a result, computational models are required to predict interactions involving entirely novel targets and disease contexts, thereby significantly increasing the complexity of the task and posing a greater challenge to their generalization capabilities.

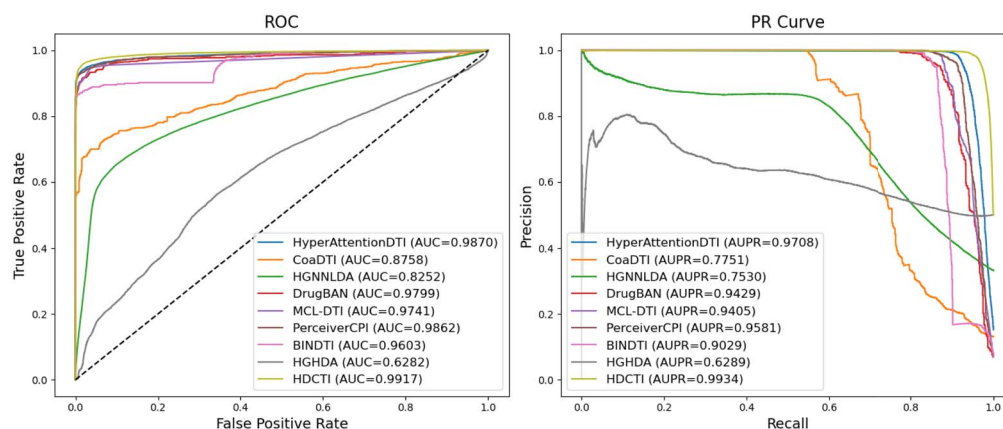
Despite the increased difficulty posed by this validation setting, HDCTI exhibits superior generalization capability. This advantage arises from its hypergraph-based framework, which explicitly incorporates disease context by modeling the complex many-to-many relationships among compounds, targets, and diseases. By leveraging hypergraph construction and disease-aware representation learning, HDCTI effectively captures higher-order semantic dependencies that are essential for predicting novel interactions beyond the training distribution. In contrast, most baseline models rely solely on observed CTIs without integrating disease and herbal information, thereby limiting their capacity to generalize under disease-aware scenarios.

Furthermore, as presented in Table 3, HDCTI consistently achieves either the best or second-best performance across key evaluation metrics, including AUC, AUPR, and F1-score. Notably, on the TCMSP dataset, HDCTI achieves a 30% improvement in AUC over the strongest baseline, HyperAttentionDTI. This substantial gain underscores the robustness of HDCTI under stringent evaluation conditions that more accurately reflect real-world scenarios, particularly those involving novel or emerging diseases.

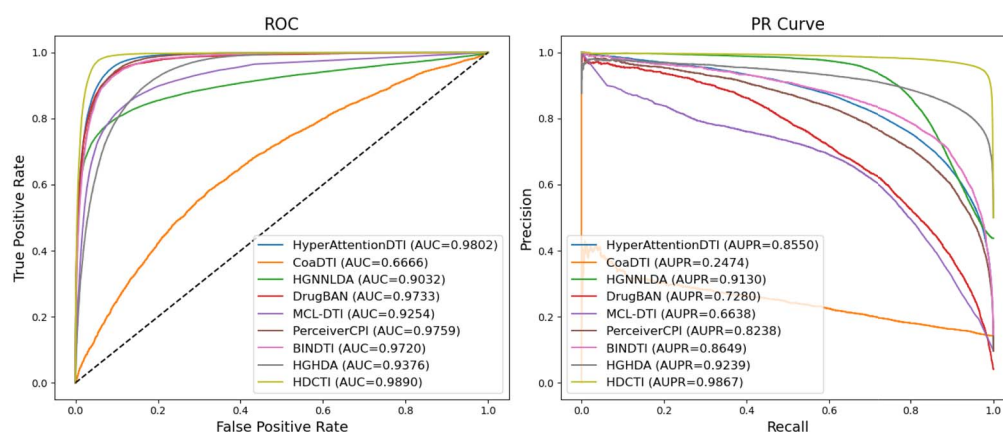
In addition to the disease-aware strategy based on randomly selecting 10 diseases, we have introduced a cluster-based cross-validation approach to improve the reproducibility and rigor of disease-aware evaluation. Specifically, using the diseases from the TCM-Suite dataset as an example, we first transformed the disease names into Term Frequency-Inverse Document Frequency (TF-IDF) vectors to extract meaningful textual features. We then applied the K-Means clustering algorithm to partition the diseases into five distinct, non-overlapping groups, which were subsequently used as folds in a five-fold cross-validation framework. Importantly, we chose to cluster diseases based on their textual names rather than their CTI profiles to prevent potential information leakage in the disease-aware validation process.

To ensure strict independence during evaluation, all CTIs associated with the diseases in the test fold were exclusively used as the testing data in each round. This rigorous setup guarantees that no disease-related information, neither at the disease nor the target level, is shared between the training and test folds. Consequently, the evaluation setting more closely simulates real-world scenarios, where predictive models must generalize to previously unseen diseases and their corresponding targets.

All baseline models and HDCTI were evaluated using the same disease-cluster-based cross-validation folds to ensure a



(a) AUC and PR curves of different methods in TCM-Suite



(b) AUC and PR curves of different methods in TCMSP

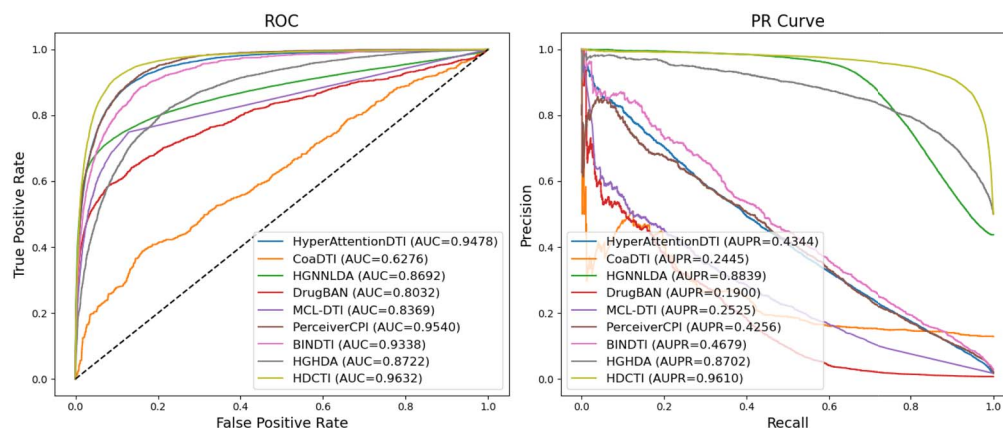


Figure 2. AUC and PR curves of different methods across three datasets: TCM-Suite, TCMSP, and SymMap2.0. Alt Text: performance curves (AUC and PR) comparing multiple methods on three datasets.

consistent and fair comparison. As shown in Table 4, HDCTI outperforms all baselines across most evaluation metrics, with particularly notable improvements in AUPR and Precision. While DrugBAN achieves slightly higher Recall in certain folds, its substantially lower Precision and F1-score indicate a higher propensity for false positive predictions. These results underscore

the robustness and generalization capability of HDCTI in challenging disease-aware scenarios.

The excellent performance of HDCTI is mainly attributed to its integration of MCMT mechanism through two independent hyper-graph neural networks. This combination significantly enhances the accuracy of CTI predictions. In summary, HDCTI is an

Table 3. Performance of different methods under disease-aware validation strategy

Dataset	Method	AUC	AUPR	Recall	Precision	F1-score
TCM-Suite	HyperAttentionDTI	0.5569(±0.0221)	0.3906(±0.0466)	0.5746(±0.3522)	0.3655(±0.0293)	0.4072(±0.1363)
	CoaDTI	0.7152(±0.0477)	0.6529(±0.0231)	0.4391(±0.2511)	0.8481(±0.1275)	0.5352(±0.1595)
	HGNNLDA	0.2060(±0.0350)	0.3746(±0.0164)	0.0319(±0.0078)	0.5681(±0.0901)	0.0601(±0.0132)
	DrugBAN	0.9665(±0.0071)	0.9599(±0.0017)	0.8862(±0.0250)	0.9286(±0.0395)	0.9062(±0.0101)
	MCL-DTI	0.9604(±0.0039)	0.9496(±0.0033)	0.8195(±0.0025)	0.9893(±0.0001)	0.8950(±0.0021)
	PerceiverCPI	0.9703(±0.0059)	0.9644(±0.0079)	0.8331(±0.0284)	0.9233(±0.0106)	0.8755(±0.0110)
	BINDTI	0.8913(±0.0110)	0.7857(±0.0197)	0.9531(±0.0297)	0.7966(±0.0143)	0.8675(±0.0059)
	Ours	0.9734(±0.0023)	0.9784(±0.0019)	0.9045(±0.0077)	0.9776(±0.0035)	0.9396(±0.0029)
	Ours	0.9734(±0.0023)	0.9784(±0.0019)	0.9045(±0.0077)	0.9776(±0.0035)	0.9396(±0.0029)
TCMSP	HyperAttentionDTI	0.7419(±0.0427)	0.6644(±0.0468)	0.5048(±0.0588)	0.6997(±0.0439)	0.5830(±0.0243)
	CoaDTI	0.6470(±0.0200)	0.1514(±0.0949)	0.0101 (±0.0050)	0.1282(±0.0527)	0.0187(±0.0111)
	HGNNLDA	0.4479(±0.0245)	0.4533(±0.0262)	0.1806(±0.0310)	0.3803(±0.0510)	0.2449(±0.0390)
	DrugBAN	0.4825(±0.1526)	0.1365(±0.1812)	0.8225(±0.1543)	0.1347(±0.1724)	0.1921(±0.1928)
	MCL-DTI	0.7400(±0.0140)	0.4628(±0.0229)	0.0785(±0.0005)	0.6576(±0.0206)	0.1402(±0.0002)
	PerceiverCPI	0.5780(±0.2164)	0.1370(±0.1546)	0.0117(±0.0090)	0.0078(±0.0067)	0.0092(±0.0074)
	BINDTI	0.5308(±0.2103)	0.1598(±0.2008)	0.9377(±0.0591)	0.1397(±0.1602)	0.7029(±0.0405)
	Ours	0.9670(±0.0001)	0.9649(±0.0004)	0.9416(±0.0011)	0.9113(±0.0030)	0.9262(±0.0010)
	Ours	0.9670(±0.0001)	0.9649(±0.0004)	0.9416(±0.0011)	0.9113(±0.0030)	0.9262(±0.0010)
SymMap2.0	HyperAttentionDTI	0.4150(±0.0714)	0.3914(±0.0201)	0.2909(±0.1123)	0.5540(±0.0976)	0.3613(±0.0896)
	CoaDTI	0.5686(±0.0277)	0.5785(±0.0334)	0.6806 (±0.0644)	0.5302 (±0.0150)	0.5947 (±0.0204)
	HGNNLDA	0.2121(±0.0088)	0.3509(±0.0027)	0.0257(±0.0026)	0.1923(±0.0120)	0.0452(±0.0040)
	DrugBAN	0.7213(±0.0224)	0.1502(±0.0214)	0.5208(±0.0595)	0.1110(±0.0217)	0.1822(±0.0307)
	MCL-DTI	0.7414(±0.0381)	0.1512(±0.0684)	0.0158(±0.0120)	0.3223(±0.1353)	0.0249(±0.0235)
	PerceiverCPI	0.7961(±0.0242)	0.1586(±0.0123)	0.0045(±0.0063)	0.4743(±0.1198)	0.0089(±0.0122)
	BINDTI	0.5601(±0.0247)	0.0533(±0.0123)	0.9443(±0.0681)	0.0510(±0.0183)	0.6775(±0.0093)
	Ours	0.9048(±0.0020)	0.9161(±0.0014)	0.8558(±0.0011)	0.8077(±0.0078)	0.8310(±0.0040)
	Ours	0.9048(±0.0020)	0.9161(±0.0014)	0.8558(±0.0011)	0.8077(±0.0078)	0.8310(±0.0040)

Note: Best results are bolded.

efficient CTI prediction method that effectively captures high-order information and key features of components and targets, thereby improving performance and surpassing other advanced models. Our experimental results further demonstrate HDCTI's ability to achieve excellent results by effectively utilizing hypergraph prediction.

Ablation study

To conduct an in-depth study on the impact of various modules of the HDCTI model on its performance, this paper investigates five variants of the model. Their detailed descriptions are as follows:

- HDCTI-n: in this variant, HDCTI employs average pooling to aggregate embeddings of compounds and target nodes from different layers of the convolutional network, instead of summing them as in the original model.
- HDCTI-x: in this variant, HDCTI uses max pooling to aggregate embeddings of compounds and target nodes from different layers of the convolutional network, instead of summing them as in the original model.
- HDCTI-a: in this variant, HDCTI removes the multi-head attention module. Embeddings of compounds and target nodes are obtained by directly using convolutional methods to gather information from neighboring nodes, rather than using multi-head attention.
- HDCTI-g: in this variant, HDCTI eliminates the self-gating mechanism.
- HDCTI-p: in this variant, HDCTI eliminates the Page-Rank Integration.

Next, these variants are compared with the performance of the original HDCTI model. Experimental results are presented in Table 5. Below are the comparative results of this study.

From the results in the table, it is evident that from the perspective of pooling, average pooling slightly outperforms max

pooling, but both are inferior to the summation pooling we utilized. Although average pooling and max pooling show relatively better performance in recall, they are accompanied by comparatively poorer precision, and their F1-score, which integrates both recall and precision, also falls short of summation pooling. This suggests that average pooling and max pooling may lead to the loss of unique features and information, thereby negatively impacting predictive performance, highlighting the effectiveness of summation pooling.

While HDCTI-a achieves AUC and AUPR close to the original model on the TCM-Suite and TCMSP datasets, its precision and F1-score have declined, especially on the SymMap2.0 dataset. This indicates that the multi-head attention mechanism plays a crucial role in extracting effective features. HDCTI-g performs slightly worse than the original model across the three datasets, but slightly better than other variants. Particularly on the SymMap2.0 dataset, HDCTI-g's performance is only slightly lower than the original model. This demonstrates the contribution of the self-gating mechanism to improving model performance. In addition, for HDCTI-p, except for the variant of PageRank fusion mechanism, we can observe that it has a certain impact on the model's performance, particularly in terms of the F1-score. This finding confirms that PageRank plays a crucial role in enhancing node embeddings through global influence propagation, thereby improving the overall predictive performance of our model.

Overall, the design of the HDCTI original model is the most reasonable and effective, with overall performance better than various variant models on different datasets, proving that the key mechanisms adopted are crucial for improving model performance.

Cross-dataset generalization evaluation

To further evaluate the predictive performance and generalization ability of our method, we conducted cross-dataset experiments by training the model on one dataset and testing

Table 4. Performance of different methods under cluster-based cross-validation approach on TCM-Suite dataset

Fold	Method	AUC	AUPR	Recall	Precision	F1-score
1	HyperAttentionDTI	0.9382	0.9125	0.8175	0.9097	0.8608
	CoaDTI	0.5550	0.5392	0.9397	0.5126	0.6663
	HGNNLDA	0.4308	0.4682	0.1603	0.4822	0.2406
	DrugBAN	0.9445	0.8789	0.8479	0.6887	0.7600
	MCL-DTI	0.9661	0.9240	0.8509	0.9309	0.8891
	PerceiverCPI	0.9557	0.9093	0.7858	0.8762	0.8282
	BINDTI	0.8575	0.4020	0.4462	0.3013	0.8274
	Ours	0.9815	0.9590	0.7931	0.9940	0.8835
2	HyperAttentionDTI	0.9551	0.9596	0.9074	0.9396	0.9232
	CoaDTI	0.6984	0.7406	0.9834	0.5445	0.6967
	HGNNLDA	0.3762	0.4361	0.1410	0.3913	0.2074
	DrugBAN	0.9771	0.9462	0.9231	0.8788	0.9004
	MCL-DTI	0.9877	0.9569	0.8841	0.9734	0.9265
	PerceiverCPI	0.9644	0.9566	0.8764	0.9085	0.8922
	BINDTI	0.9360	0.6428	0.5988	0.4449	0.9219
	Ours	0.9735	0.9782	0.8839	0.9393	0.9108
3	HyperAttentionDTI	0.9492	0.9626	0.8991	0.9299	0.9142
	CoaDTI	0.7348	0.7466	0.7936	0.4115	0.5403
	HGNNLDA	0.3347	0.4175	0.1332	0.3444	0.1921
	DrugBAN	0.9839	0.9512	0.9333	0.7508	0.8322
	MCL-DTI	0.9883	0.9466	0.8889	0.9558	0.9211
	PerceiverCPI	0.9327	0.9408	0.8536	0.8257	0.8394
	BINDTI	0.9331	0.5231	0.6309	0.4740	0.9374
	Ours	0.9744	0.9800	0.8851	0.9455	0.9134
4	HyperAttentionDTI	0.9396	0.9487	0.8758	0.9153	0.8951
	CoaDTI	0.6992	0.7331	0.7058	0.5714	0.6317
	HGNNLDA	0.3914	0.4617	0.1782	0.4050	0.2475
	DrugBAN	0.9800	0.9423	0.9268	0.7755	0.8444
	MCL-DTI	0.9815	0.9289	0.8739	0.9123	0.8927
	PerceiverCPI	0.9617	0.8897	0.8088	0.8999	0.8516
	BINDTI	0.9151	0.5818	0.5655	0.4109	0.9013
	Ours	0.9736	0.9798	0.8841	0.9410	0.9117
5	HyperAttentionDTI	0.9459	0.9012	0.8353	0.9122	0.8722
	CoaDTI	0.6716	0.7358	0.8421	0.2388	0.3710
	HGNNLDA	0.3273	0.4231	0.1581	0.3592	0.2195
	DrugBAN	0.9451	0.8612	0.8382	0.6552	0.7355
	MCL-DTI	0.9507	0.8685	0.8060	0.9474	0.8709
	PerceiverCPI	0.9531	0.9233	0.8319	0.8795	0.8555
	BINDTI	0.8944	0.5388	0.4395	0.2957	0.8636
	Ours	0.9730	0.9797	0.8818	0.9433	0.9115
Mean±SD	HyperAttentionDTI	0.9456(±0.0004)	0.9369(±0.0005)	0.8670(±0.0013)	0.9213(±0.0001)	0.8931(±0.0005)
	CoaDTI	0.6718(±0.0041)	0.6991(±0.0077)	0.8529(±0.0123)	0.4558(±0.0150)	0.5812(±0.0141)
	HGNNLDA	0.3720(±0.0016)	0.4413(±0.0004)	0.1541(±0.0003)	0.3964(±0.0020)	0.2221(±0.0005)
	DrugBAN	0.9661(±0.0004)	0.9159(±0.0013)	0.8938(±0.0011)	0.7498(±0.0061)	0.8145(±0.0031)
	MCL-DTI	0.9748(±0.0002)	0.9249(±0.0010)	0.8607(±0.0010)	0.9439(±0.0006)	0.9000(±0.0006)
	PerceiverCPI	0.9535(±0.0001)	0.9239(±0.0005)	0.8313(±0.0011)	0.8770(±0.0007)	0.8533(±0.0005)
	BINDTI	0.9072(±0.0008)	0.5377(±0.0070)	0.5362(±0.0062)	0.3853(±0.0051)	0.8903(±0.0015)
	Ours	0.9752(±0.0029)	0.9753(±0.0076)	0.8656(±0.0378)	0.9526(±0.0211)	0.9061(±0.0113)

The best scores are bolded

it on another. Specifically, we selected three datasets—TCM Suite, TCMSp, and SymMap2.0—and systematically designed experiments where the model was trained on one dataset and used to make predictions on the other two. This process was repeated for all possible training-testing combinations to comprehensively assess the model’s generalization capability across different datasets. To highlight the superiority of our approach, we compared it with all baseline methods from the cross-validation experiments, as well as models from the ablation study. AUC was used as the evaluation metric, and the results are summarized in Table 6.

The experimental results indicate that traditional baseline models, such as HyperAttentionDTI and PreceiverCPI, perform

poorly in cross-dataset testing, with AUC values close to random levels, demonstrating their limited generalization ability. In contrast, the HDCTI series models achieve significantly higher AUC values, suggesting that the introduced improvements enhance adaptability across different datasets. Among them, HDCTI-g performs well in multiple train–test combinations, while the final HDCTI model achieves the highest AUC values in all experiments, fully demonstrating its cross-dataset generalization capability. However, some fluctuations in prediction performance across datasets can still be observed, such as slightly lower AUC values for SymMap2.0, which may be attributed to differences in dataset distributions. This suggests that future research could further optimize the model to enhance its robustness across datasets.

Table 5. Performance comparison of different variants on three datasets

Dataset	Method	AUC	AUPR	Recall	Precision	F1-score
TCM-Suite	HDCTI-n	0.9884(±0.0008)	0.9907(±0.0005)	0.9568(±0.0020)	0.9582(±0.0079)	0.9575(±0.0042)
	HDCTI-x	0.9765(±0.0032)	0.9843(±0.0018)	0.9558(±0.0038)	0.9286(±0.0067)	0.9420(±0.0017)
	HDCTI-a	0.9901(±0.0013)	0.9923(±0.0010)	0.9474(±0.0026)	0.9863(±0.0022)	0.9665(±0.0007)
	HDCTI-g	0.9862(±0.0009)	0.9894(±0.0006)	0.9465(±0.0024)	0.9437(±0.0025)	0.9451(±0.0018)
	HDCTI-p	0.9914(±0.0005)	0.9929(±0.0005)	0.9505(±0.0022)	0.9826(±0.0039)	0.9663(±0.0008)
	HDCTI	0.9917(±0.0005)	0.9934(±0.0002)	0.9475(±0.0014)	0.9887(±0.0014)	0.9677(±0.0008)
TCMSP	HDCTI-n	0.9874(±0.0003)	0.9843(±0.0004)	0.9799(±0.0011)	0.9382(±0.0010)	0.9586(±0.0002)
	HDCTI-x	0.9801(±0.0016)	0.9787(±0.0015)	0.9791(±0.0008)	0.8848(±0.0058)	0.9296(±0.0033)
	HDCTI-a	0.9788(±0.0003)	0.9749(±0.0009)	0.9636(±0.0016)	0.9315(±0.0024)	0.9473(±0.0013)
	HDCTI-g	0.9886(±0.0007)	0.9860(±0.0012)	0.9793(±0.0011)	0.9406(±0.0022)	0.9596(±0.0013)
	HDCTI-p	0.9885(±0.0004)	0.9862(±0.0006)	0.9758(±0.0019)	0.9435(±0.0007)	0.9594(±0.0008)
	HDCTI	0.9890(±0.0005)	0.9867(±0.0007)	0.9781(±0.0011)	0.9439(±0.0016)	0.9607(±0.0011)
SymMap2.0	HDCTI-n	0.9613(±0.0009)	0.9602(±0.0018)	0.9192(±0.0028)	0.8929(±0.0024)	0.9059(±0.0015)
	HDCTI-x	0.9519(±0.0017)	0.9512(±0.0021)	0.9578(±0.0015)	0.8118(±0.0038)	0.8788(±0.0028)
	HDCTI-a	0.9604(±0.0021)	0.9587(±0.0022)	0.9136(±0.0026)	0.8975(±0.0051)	0.9055(±0.0035)
	HDCTI-g	0.9624(±0.0010)	0.9610(±0.0021)	0.9184(±0.0012)	0.8969(±0.0019)	0.9075(±0.0009)
	HDCTI-p	0.9611(±0.0010)	0.9583(±0.0009)	0.9041(±0.0055)	0.9000(±0.0026)	0.9020(±0.0037)
	HDCTI	0.9632(±0.0010)	0.9610(±0.0015)	0.9180(±0.0019)	0.8979(±0.0028)	0.9078(±0.0017)

Note: Best results are bolded.

Table 6. AUC Performance of different models on cross-dataset validation

	TCM-Suite		TCMSP		SymMap2.0	
	TCMSP	SymMap2.0	TCM-Suite	SymMap2.0	TCM-Suite	TCMSP
HyperAttentionDTI	0.5038	0.5389	0.4849	0.5248	0.5514	0.4391
CoaDTI	0.5425	0.6117	0.4824	0.4309	0.5419	0.5370
HGNNLDA	0.5054	0.5515	0.5196	0.4985	0.5432	0.5008
DrugBAN	0.6059	0.5189	0.6301	0.5159	0.6078	0.4657
MCL-DTI	0.5089	0.5325	0.4944	0.5260	0.5507	0.4069
PreceiverCPI	0.5064	0.5004	0.5221	0.5222	0.5774	0.4223
BINDTI	0.4798	0.4953	0.5232	0.4790	0.5418	0.5436
HDCTI-n	0.8735	0.7518	0.8240	0.8442	0.8396	0.7920
HDCTI-x	0.8652	0.8173	0.7876	0.5222	0.5090	0.5607
HDCTI-a	0.8503	0.6711	0.8682	0.7573	0.7622	0.7854
HDCTI-g	0.9047	0.8557	0.8764	0.9291	0.9018	0.8899
HDCTI-p	0.8085	0.7743	0.7185	0.8882	0.7219	0.7076
HDCTI	0.9239	0.9147	0.9305	0.9382	0.9247	0.9195

The best scores are bolded

Overall, these experimental results confirm the superior performance of HDCTI in handling distributional differences, providing strong support for its practical application in real-world scenarios.

The impact of network layers on model performance

We investigated the impact of different network layers on HDCTI performance by conducting a series of experiments. The evaluation was based on AUC, AUPR, and F1-score. The results are shown in Fig. 3.

When the number of network layers is no more than three, the model's performance remains relatively stable across the three datasets. Specifically, in the TCM-Suite dataset, the model achieves the best overall performance with two layers. In the TCMSP and SymMap2.0 datasets, the model performs better with one layer compared with two layers, but the best performance is achieved with three layers. However, when the number of layers exceeds three, the model's performance decreases across all three datasets. This suggests that when the model aggregates higher-order information in the hypergraph, it may suffer from over-smoothing, leading to a decline in performance.

Case analysis

In our case study, we applied the HDCTI model to reposition two natural compounds: coumarin and progesterone. Initially, we trained HDCTI using all known CTIs while excluding those involving coumarin and progesterone. We then used the trained model to predict the associations between these compounds and their potential targets. The targets were ranked based on prediction scores, with details available in Tables 7 and 8. Notably, among the Top-10 predicted targets for coumarin and progesterone, 7 and 8 targets, respectively, were validated in the existing literature. To further emphasize the effectiveness of our model, we applied the same experimental setup to all baseline methods. For the two natural compounds in our case study, the Top-10 predicted targets of the baseline methods are shown in Table 9. Taking Coumarin as an illustrative example, none of the targets predicted by CoaDTI and HGNNLDA are validated in the PubChem database. Even the second-best performing models, namely HyperAttentionDTI and Preceiver, each yield only four validated targets. In contrast, our proposed method successfully identifies seven validated targets, demonstrating its superior predictive accuracy and practical utility. Upon further analysis, we observed that none

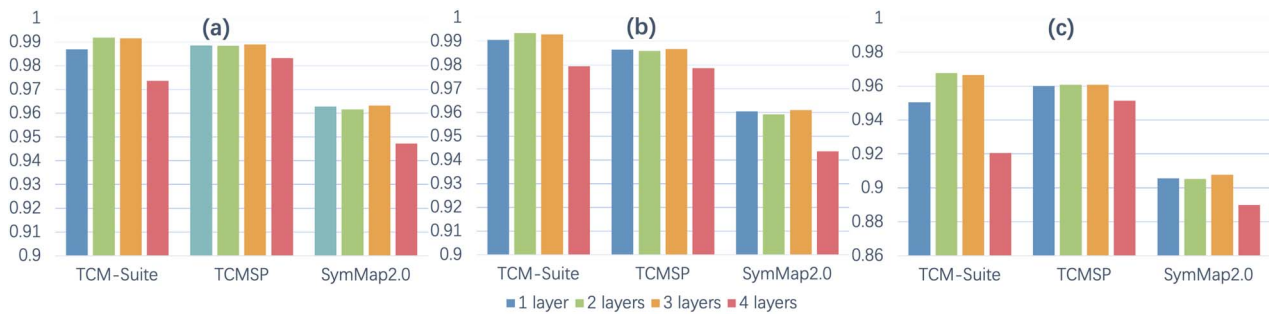


Figure 3. The results of different network layers on model performance. (a) The performance of AUC, (b) the performance of AUPR, and (c) the performance of F1-score. Alt Text: Bar charts showing how the number of network layers affects AUC, AUPR, and F1-score metrics.

Table 7. Top-10 predicted targets for Coumarin

Rank	Protein name	Score	Evidence
1	CYP2E1	0.9536	PMID:34334558
2	AR	0.9530	N/A
3	CYP3A4	0.9426	DGIdb
4	CPA6	0.9414	N/A
5	PARP1	0.9369	PMID:31199928
6	NOS2	0.9357	PMID:32578917
7	CASP3	0.9339	PMID:32578917
8	CASP8	0.9329	PMID:32578917
9	TNF	0.9316	PMID:32578917
10	ABCB1	0.9313	N/A

Note: N/A indicates that no relevant literature was found.

Table 8. Top-10 predicted targets for Progesterone

Rank	Protein name	Score	Evidence
1	PDE10A	0.9985	N/A
2	NFKBIA	0.9978	PMID: 17437195
3	CASP3	0.9976	PMID: 21933904
4	AKT1	0.9975	PMID: 15358673
5	MMP9	0.9972	PMID: 17404688
6	EDN1	0.9970	PMID: 21795739
7	CYP1A1	0.9968	PMID: 9667077
8	IL4	0.9967	PMID: 22615136
9	BAX	0.9967	PMID: 20876846
10	PDE4A	0.9965	N/A

Note: N/A indicates that no relevant literature was found.

of the validated targets identified by the baseline models ranked within HDCTI's Top-10 predictions, reflecting the differences in prioritization among the models. The predicted scores for these targets are reported in Table 10. It is noteworthy that HDCTI assigned scores exceeding 0.5 to all these targets, indicating that it also recognized their potential associations, but with slightly lower confidence compared with its top-ranked candidates. Further discussion is provided below.

Coumarin is a natural compound widely present in various CHMs, such as licorice, celery, and Astragalus [35]. It exhibits several pharmacological effects, including anticoagulant, anti-inflammatory, and antioxidant properties [36]. Furthermore, coumarin has demonstrated significant potential in metabolic regulation, showing promise for improving insulin resistance and treating type 2 diabetes due to its strong anti-inflammatory and antioxidant capabilities [37]. While there is currently no direct evidence that coumarin can bind to the androgen receptor (AR), its structural similarity to chrysin, a natural compound found in plantain, suggests this possibility. As shown in Fig. 4, the active

Table 9. The number of targets verified in Pubchem among the Top-10 prediction results for two compounds by each baseline method

Method	Coumarin	Progesterone
HyperAttentionDTI	4	3
CoaDTI	0	2
HGNNLDA	0	5
DrugBAN	3	6
MCL-DTI	2	5
PreceiverCPI	4	3
BINDTI	3	4
HDCTI	7	8

components of Astragalus and Plantain and their target effects in treating type 2 diabetes are illustrated. In the figure, dashed lines represent the predicted interactions of coumarin with AR based on HDCTI, while solid lines indicate relationships present in the dataset used to train HDCTI. Chrysin's chemical structure, which includes an aromatic ring and a benzopyran structure formed by an oxygen bridge linking the central pyran ring, closely resembles that of coumarin [38]. Due to these shared structural characteristics, HDCTI can infer a potential interaction between coumarin and AR. During the learning process, HDCTI highlights coumarin's structural similarity to chrysin, thus supporting the likelihood of similar binding behavior. HDCTI leverages this structural similarity in its predictive process for two primary reasons. First, through hypergraph convolution and multi-head attention mechanisms, HDCTI captures higher-order associations and network contexts among compounds and targets. This allows HDCTI to learn not only from direct interactions but also from indirect patterns and shared structural or functional properties among compounds. Second, by incorporating prior knowledge of compound-protein interactions and structural features, HDCTI enhances its ability to predict interactions based on compound similarity, inferring that coumarin, like chrysin, may interact with AR.

Progesterone is a crucial hormone known to regulate the neurotransmitter systems related to dopamine and glutamate. It specifically influences dopamine synthesis and release in certain brain regions, affecting both the availability and signaling pathways of this neurotransmitter [39, 40]. Additionally, progesterone modulates dopamine signaling by altering receptor sensitivity or by adjusting the number of dopamine receptors present [41]. Beyond its effects on dopamine, progesterone also plays a key role in regulating glutamate signaling, particularly by modulating the activity of N-methyl-d-aspartate receptors, which are central to neuronal excitability and synaptic plasticity [42]. This connection

Table 10. Prediction scores of HDCTI for the PubChem-validated targets identified by baselines

Compound	Target	Score	Compound	Target	Score
Coumarin	CYP2C9	0.846	Progesterone	NR3C1	0.5943
Coumarin	DMD	0.9033	Progesterone	PTGS1	0.9035
Coumarin	HDAC7	0.8669	Progesterone	F2	0.7202
Coumarin	CAT	0.5617	Progesterone	CA2	0.732
Coumarin	PTGS2	0.8938	Progesterone	GABRA2	0.816
Coumarin	NR3C1	0.9251	Progesterone	TOP2A	0.7197
Coumarin	HSP90AA1	0.8314	Progesterone	DPP4	0.8094
Coumarin	GABRA2	0.8994	Progesterone	AR	0.8681
Coumarin	PTGS1	0.8899	Progesterone	TGFBR2	0.7216
Coumarin	GABRA5	0.7375	Progesterone	TLR4	0.77
Coumarin	HTT	0.921	Progesterone	IGF1	0.5991
Coumarin	DTL	0.8602	Progesterone	ASNS	0.516
Coumarin	MCM7	0.8861	Progesterone	ARF6	0.9307
Progesterone	F10	0.8593	Progesterone	IL2	0.8629
Progesterone	MMP1	0.8667	Progesterone	AKR1C3	0.6487
Progesterone	GSTP1	0.9457	Progesterone	CYP2C9	0.9871
Progesterone	IL1B	0.93			

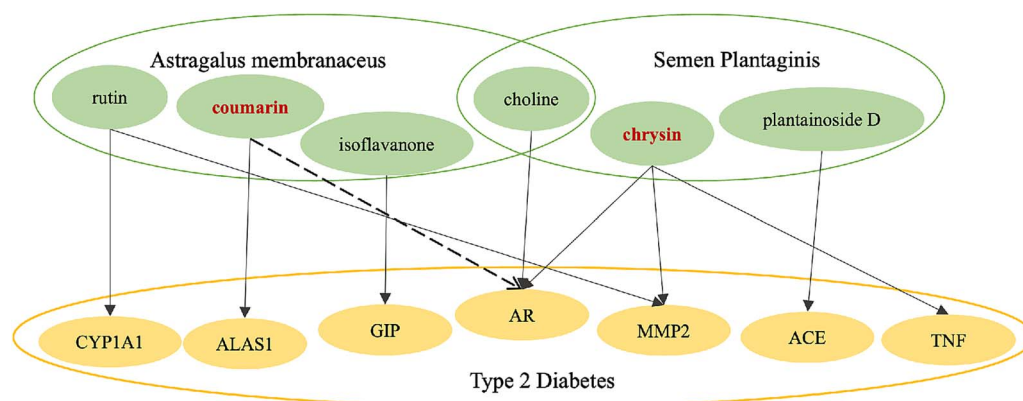


Figure 4. Network topology of astragalus and plantain components targeting type 2 diabetes. Alt Text: A CTI network graph connecting components of astragalus and plantain with diabetes-related targets.

between progesterone and neurotransmitter systems aligns with the potential interaction predicted by our HDCTI model between progesterone and PDE10A with the highest score. The PDE10A gene encodes a phosphodiesterase that primarily degrades cyclic adenosine monophosphate (cAMP) and cyclic guanosine monophosphate (cGMP), which are essential to dopamine and glutamate signaling pathways [43]. By controlling the degradation rate of cAMP, PDE10A precisely regulates dopamine signaling duration and intensity [44]. Additionally, PDE10A influences glutamate-related processes, such as long-term potentiation and long-term depression, by modulating cGMP levels [45]. Since progesterone is known to impact dopamine and glutamate signaling, which are regulated by PDE10A through cAMP and cGMP degradation, the HDCTI model's prediction of a CTI between progesterone and PDE10A is strongly supported. This pathway overlap suggests a potential mechanism that warrants further validation studies to confirm the exact nature of this interaction.

Our experimental results suggest that coumarin may offer novel insights into type 2 diabetes treatment through its potential impact on the AR signaling pathway. By influencing AR signaling, coumarin could contribute to innovative approaches for managing metabolic disorders. Similarly, progesterone's role in modulating dopamine and glutamate signaling opens promising avenues for research into neurological disorders. Through its effects on

these neurotransmitter systems, progesterone holds potential for developing therapeutic strategies for nervous system-related diseases. Our case studies demonstrate that HDCTI effectively identifies target sites for active herbal compounds, with validation from relevant literature supporting the model's reliability and practicality. Consequently, HDCTI proves to be a valuable tool for predicting potential CTIs and guiding research in traditional medicine. This model provides substantial support for exploring the therapeutic potential of CHM.

Conclusion

In this work, we present HDCTI as a powerful approach for predicting CTIs in complex TCM systems through hypergraph representation learning. By modeling the MCMT relationships as hypergraphs, HDCTI leverages hypergraph convolution and multi-head attention mechanisms to capture intricate interaction patterns among herbal compounds and disease targets. This approach enables the model to generate embeddings for compounds and targets to predict potential CTIs in a seamless, end-to-end manner. Our experimental results demonstrate HDCTI's superior performance over existing methods, showcasing its potential to advance TCM research and facilitate the discovery of new therapeutic targets. Regarding future work, the integration

of additional biological data, such as compound SMILES and target sequence information, and the exploration of advanced optimization techniques will be crucial to further enhance HDCTI's accuracy and expand its utility in herbal medicine repositioning. Overall, HDCTI stands as a valuable tool for uncovering the therapeutic potential of traditional herbal compounds within the framework of modern pharmacology.

Key Points

- We present a novel hypergraph representation learning framework, hypergraph representation learning for compound–target interaction (HDCTI), designed to accurately identify therapeutic targets of natural compounds in traditional Chinese medicine (TCM) herbs. By leveraging hypergraph representation learning, HDCTI effectively captures the intricate multi-component, multi-target characteristics inherent in TCM. This capability enables it to address higher-order relationships and synergistic effects that are beyond the scope of existing single-component, single-target prediction methods.
- HDCTI constructs two distinct hypergraphs: one representing herb–compound relationships and the other capturing disease–target associations. Through the integration of multi-head attention and hypergraph convolution, HDCTI refines embeddings to uncover higher-order interactions, improving the accuracy of compound–target interaction (CTI) predictions.
- Experimental results demonstrate that HDCTI outperforms existing CTI prediction models across multiple datasets. Its efficacy is further validated in case studies involving compounds like coumarin and progesterone, showcasing its ability to predict potential therapeutic targets within complex network pharmacology frameworks.

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Author contributions

Conceptualization and study design: Yantong Qiao and Lun Hu; performing experiments and statistical analysis: Yantong Qiao and Pengwei Hu; data processing and integration: Yantong Qiao and Jun Zhang; manuscript preparation: Yantong Qiao and Lun Hu; manuscript editing and review: Lun Hu and Xin Luo.

Conflict of interests

None declared.

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Data availability

The dataset and source code can be freely downloaded from <https://github.com/tong87-bio/HDCTI>.

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